Anneru Global Energy Hackathon 2026 · Availability · Digital Solutions, Apps & AI/ML Tools · Non-residential (healthcare)

Hospital Uptime

Tiered solar, battery and generator plans that keep a hospital's life-critical loads on through 8 to 20 hours of daily grid outage.

Problem

Nigerian hospitals lose grid power 8-20 hours a day. Most run every load on one diesel genset for every outage hour: high fuel cost, dependence on fuel supply, and a single point of failure for oxygen, vaccine fridges and ICU equipment.

Solution

- Equipment is listed by tier: 1 life-critical (never shed), 2 essential, 3 comfort (shed first, never on diesel).
- The app sizes solar, battery and a small generator for tiers 1-2, simulates every hour, stress-tests 8-20 h outages on sunny and cloudy days, and shows cost, monthly payment, diesel saved and payback.
- The printed plan goes to an installer for quotes and to a lender for financing.

Results (sample hospital, 14 h outage)

Life-critical unserved	Diesel	Payback
0 kWh	**21 → 2.5 L/day**	**4.3 years**

All 8 stress cases keep tiers 1-2 on; diesel 0-4.4 L/day; only comfort load is shed.



Next 6-18 months

- Pilot with one hospital using real meter data.
- Installer quotes in the cost model; lender-ready export.
- Inputs (outage, sun, prices) make it portable to any region.

Team and disclosure

Alan De Bortolo, founder, Design Master Solutions LLC (solo). Repository started Oct 29, 2026, 22:06 EDT. Built with Claude Code as coding assistant; no third-party code or datasets. Prices are editable planning assumptions, not quotes.

Live: hospital-uptime.designmaster.workers.dev · Code (MIT): github.com/DesignMaster-Solutions/hospital-uptime · Demo: youtu.be/Y5FrlzyPXL0